DIPLOMA PROGRAMME PRACTICE / HL / SET A

Economics / Paper 3

Worked mark scheme · 105 minutes · 60 marks

Answer all parts of both questions. A calculator is permitted. Write your answers and diagrams in the spaces provided. Show working and units. Give exact values or round to two decimal places unless otherwise instructed.

Judge extended responses as a whole using the markband for that part. Award valid alternative analysis and diagrams. In linked calculations, credit correct use of an earlier result. The suggested judgements are examples.

HOW TO MARK THIS PAPER

Use the analytic scheme in part (a). Apply the 10-mark policy scale to each part (b). Add the marks for both compulsory questions.

10-MARK SCALE

0: no relevant response. 1–2: a policy is named with minimal reasoning. 3–4: a plausible policy with partial explanation. 5–6: a relevant mechanism and some use of the case, with limited trade-offs. 7–8: developed theory, relevant data and reasoned consideration of limitations. 9–10: a well-supported, feasible recommendation synthesising case evidence, economic reasoning and the most consequential trade-offs. Credit a different justified recommendation.

QUESTIONS IN THIS SCHEME

1 A levy on single-use containers [30]

2 An output gap and a repair programme [30]

QUESTION 1 / 30 MARKS

A levy on single-use containers

PART a(i) / 2 MARKS

Calculate the equilibrium price before intervention.

Working. Set 120 − 2P = 20 + 2P (1). P = 25 crowns per container (1).

PART a(ii) / 2 MARKS

Calculate equilibrium quantity and total external damage before intervention.

Working. At P = 25, quantity = 70 thousand per week (1). Damage = 70,000 × 6 = 420,000 crowns per week (1). This is damage, not firms' production cost.

PART a(iii) / 3 MARKS

Calculate the buyer's and seller's prices after the proposed tax.

Working. Pc = Pp + 10 (1). Solving 120 − 2(Pp + 10) = 20 + 2Pp gives Pp = 20 (1) and Pc = 30 crowns (1).

PART a(iv) / 3 MARKS

Calculate the new weekly quantity and tax revenue. State the share of the per-unit levy borne by buyers.

Working. Quantity = 120 − 2 × 30 = 60 thousand (1). Revenue = 10 × 60,000 = 600,000 crowns per week (1). Buyers pay 5 more than before, or half the levy (1).

PART a(v) / 2 MARKS

Calculate the weekly external damage avoided under the proposed levy.

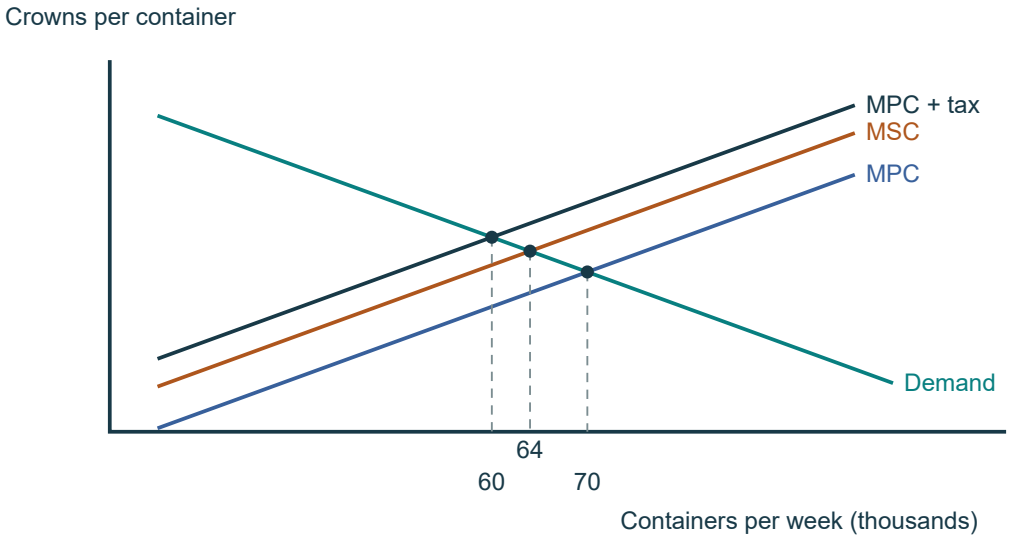
Working. Avoided output = (70 − 60) thousand (1). Avoided damage = 10,000 × 6 = 60,000 crowns per week (1).

PART a(vi) / 4 MARKS

Using a diagram, explain why the proposed levy could reduce quantity below the socially efficient level.

Diagram and explanation. Draw demand, MPC and MSC, with MSC 6 crowns above MPC. Mark Q* where demand meets MSC. With Pc − Pp = 6, Q* = 64 thousand. The proposed 10-crown wedge produces 60 thousand. Units 60 to 64 thousand have benefit above full cost; preventing them sacrifices welfare. Up to two marks for a consistent diagram and two for explanation.

WORKED REPRESENTATION / a(vi)



PART a(vii) / 4 MARKS

Calculate the change in total weekly welfare from the proposed levy relative to no intervention. Treat tax revenue as a transfer.

Working. Lost private gains from trades no longer made form a triangle (1), ½ × 10 × 10,000 = 50,000 crowns (1). Avoided damage = 60,000 crowns (1). Net welfare improves by 10,000 crowns per week (1), despite overshooting the optimum. No second addition of revenue.

PART b / 10 MARKS

Using the information and your knowledge of economics, recommend one policy for Harbour City to reduce the social cost of disposable containers.

Worked recommendation. Introduce a levy nearer the estimated 6-crown marginal damage. It aligns the private decision with full marginal cost, reducing quantity from 70 to 64 thousand. Under the linear assumptions, the improvement relative to no intervention is ½ × 6 × 6,000 = 18,000 crowns weekly, greater than the 10,000 from the proposed rate. Revenue is a transfer but could finance washing facilities.

Trade-offs. The damage estimate may omit marine harms or vary by location. A charge per container gives a weak incentive to redesign a container if all pay the same rate. Outlets without washing space have few substitutes and may pass costs to customers. A deposit can target litter more precisely when returns are convenient but collection requires resources. Pilot the levy and measure actual disposal and substitution; accept a justified deposit recommendation.

Apply the 10-mark scale on the first page to this response as a whole.

QUESTION 2 / 30 MARKS

An output gap and a repair programme

PART a(i) / 2 MARKS

Calculate the recessionary output gap as a percentage of potential GDP.

Working. (1,000 − 940)/1,000 × 100 (1) = 6% below potential (1). Accept a signed gap of −6% if defined as actual minus potential.

PART a(ii) / 2 MARKS

Calculate the simple expenditure multiplier.

Working. Total marginal leakage = 0.20 + 0.10 + 0.20 = 0.50 (1). Multiplier = 1/0.50 = 2 (1).

PART a(iii) / 3 MARKS

Using the multiplier, calculate the predicted increase in equilibrium income and the resulting GDP if the full repair plan is an additional injection.

Working. Substitute 40 billion × 2 (1). Increase = 80 billion (1). Resulting GDP = 1,020 billion crowns at model prices (1). This is a demand-side projection.

PART a(iv) / 3 MARKS

Calculate the additional government purchases that would close the initial gap in this model and compare your result with the proposed plan.

Working. Gap is 60 billion (1). Required injection = 60/2 = 30 billion (1). The plan exceeds that injection by 10 billion crowns (1); distinguish this from the 20 billion difference in predicted income.

PART a(v) / 2 MARKS

If GDP rises by 60 billion crowns, calculate the induced increase in net tax receipts and in imports under the stated propensities.

Working. Net taxes: 0.10 × 60 = 6 billion (1). Imports: 0.20 × 60 = 12 billion crowns (1).

PART a(vi) / 4 MARKS

Assuming a 30 billion purchase, the 6 billion increase in net tax receipts and no other budget changes, calculate the new deficit and year-end debt. Explain the distinction.

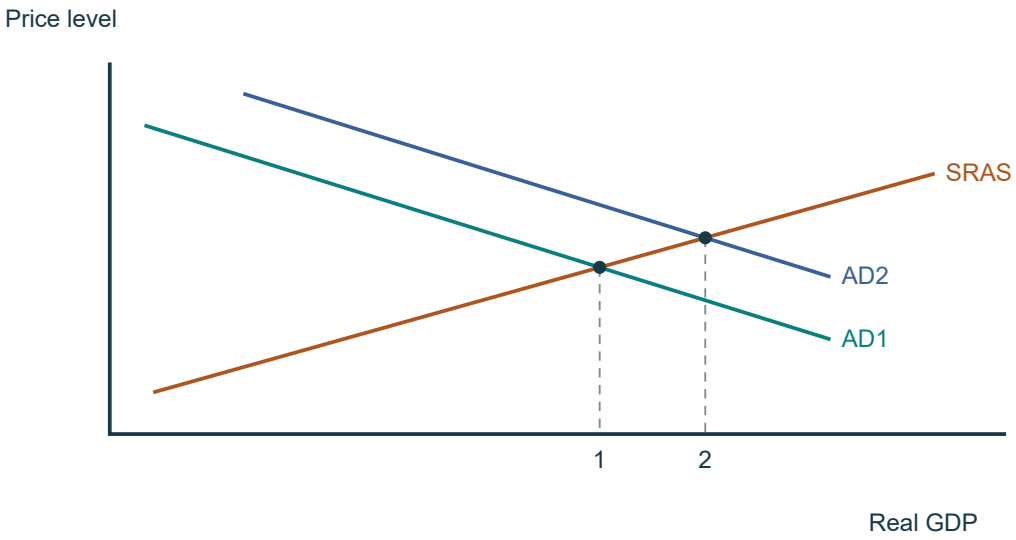
Working. New deficit = 18 + 30 − 6 = 42 billion crowns (1). If financed entirely by borrowing with no valuation changes or asset transactions, end debt = 600 + 42 = 642 billion (1). The deficit is an annual flow (1). Debt is the accumulated outstanding stock at a date (1).

PART a(vii) / 4 MARKS

Using an AD–AS diagram, explain why the repair plan might raise real GDP by less than its multiplier prediction.

Diagram and explanation. Show AD shifting right and the economy moving toward capacity on a rising SRAS, with both a higher price level and higher real output (up to 2). Sectoral shortages, rising prices or tighter interest rates can limit the real expansion. A spending multiplier does not guarantee 1,020 billion of sustainable real output (up to 2).

WORKED REPRESENTATION / a(vii)



PART b / 10 MARKS

Using the information and your knowledge of economics, recommend a fiscal response to the recessionary gap in Bracken.

Worked recommendation. Phase repairs, starting with projects using idle local labour. The model suggests an additional injection near 30 billion, while the full 40 billion would imply demand above potential. Repairs directly create orders and can raise future capacity. Prioritise projects with low import content and verifiable maintenance value rather than a fixed spending target.

Trade-offs. The multiplier is uncertain and national slack does not remove local material bottlenecks. Automatic tax receipts partly offset the cost but projected debt still rises to 642 billion. A targeted transfer is quicker yet some of the initial payment is saved; compare its effective injection rather than applying the purchases multiplier to its face value. Stage commitments against unemployment, costs and repair bids. Credit a well-supported transfer or combined implementation of one principal policy.

Apply the 10-mark scale on the first page to this response as a whole.